

Advanced Java Course Topics (Serial-Wise)

1. Introduction to Advanced Java

- Difference between Core Java and Advanced Java
- JDK, JRE, JVM recap
- Packages & API overview
- Java EE architecture

2. JDBC (Java Database Connectivity)

- JDBC Drivers (Types 1-4)
- Steps to connect Java + Database
- Statement, PreparedStatement, CallableStatement
- Batch processing
- ResultSet & ResultSetMetaData
- Transactions (commit, rollback)
- Connection Pooling
- Mini Project with JDBC

3. Servlet Programming

- Servlet Lifecycle
- HttpServlet class
- GET vs POST
- Request & Response
- RequestDispatcher (forward/include)
- init(), service(), destroy()
- Cookies
- Session Management
- ServletConfig & ServletContext
- Servlet Filters
- Error Handling in Servlets

4. JSP (Java Server Pages)

- JSP Architecture
- JSP Lifecycle
- Implicit Objects
- Scripting elements

- Directives (page, include, taglib)
- JSP Action Tags
- JSP Beans
- Exception Handling in JSP
- MVC Architecture with JSP + Servlets

5. Java Beans

- Introduction to JavaBeans
- Properties & Methods
- Introspection
- Serialization
- Using Beans in JSP

6. Hibernate (ORM Framework)

- Introduction to ORM
- Hibernate architecture
- Configuring Hibernate
- Sessions, SessionFactory
- Hibernate CRUD
- HQL
- Criteria API
- Mapping (1-1, 1-M, M-M)
- Caching
- Hibernate + MySQL Project

7. Spring Framework

- Introduction to Spring & IOC
- Dependency Injection
- Spring Beans & Bean Scopes
- Autowiring
- Spring JDBC
- Spring MVC
- Controllers, Views, and Handlers
- Form Handling
- Spring Boot Basics
- REST API CRUD Project

8. JSP + Servlet + JDBC Integration Project

- Login/Signup
- Validation
- Sessions
- Database CRUD
- MVC Implementation

9. Web Technologies

- HTML5 basics
- CSS basics
- JavaScript basics
- XML & JSON
- AJAX overview

10. Build & Deployment Tools

- Maven
- Gradle (optional)
- WAR/JAR packaging
- Deploying on Tomcat

11. Additional Concepts

- RESTful Web Services
- JSON Processing
- Microservices basics
- Logging (Log4j/SLF4J)
- Git + GitHub basics